

The empirical recurrence for OEIS A302370, recovered from its tail

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Abstract

OEIS A302370 records “Empirical recurrence of order 63” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{63} = 4956 \neq 0$, so no further tail satisfies anything shorter; any order-63 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302370 is “Number of $4 \times n$ 0..1 arrays with every element equal to 0, 2, 3 or 4 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 3, 109, 77, 918, 3125, 21831, 125193, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Apr 06 2018 12:23:45, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 63: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 63.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 4$ on, it returns order 63 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	13	c_{17}	−48048139	c_{33}	−20391333645	c_{49}	−722813920
c_2	−12	c_{18}	205565185	c_{34}	12531736990	c_{50}	228740121
c_3	−444	c_{19}	−110313196	c_{35}	14786204303	c_{51}	419256936
c_4	1306	c_{20}	−531323277	c_{36}	−15331144692	c_{52}	75484954
c_5	6125	c_{21}	830843213	c_{37}	−6387710342	c_{53}	−151097439
c_6	−26562	c_{22}	537438805	c_{38}	11652007706	c_{54}	−113724319
c_7	−39382	c_{23}	−2234953585	c_{39}	1162935471	c_{55}	2090018
c_8	270249	c_{24}	1012941221	c_{40}	−6119122487	c_{56}	34894785
c_9	59477	c_{25}	2853961445	c_{41}	−666862835	c_{57}	15361588
c_{10}	−1613653	c_{26}	−4766412568	c_{42}	3000842689	c_{58}	674026
c_{11}	924135	c_{27}	596704788	c_{43}	1711517745	c_{59}	−3145060
c_{12}	5695987	c_{28}	7971682172	c_{44}	−2232413494	c_{60}	−1670294
c_{13}	−8254234	c_{29}	−9385329115	c_{45}	−1620637960	c_{61}	114606
c_{14}	−7245251	c_{30}	−5314296089	c_{46}	1569083534	c_{62}	182468
c_{15}	31700084	c_{31}	18297000323	c_{47}	1059074858	c_{63}	4956
c_{16}	−32996121	c_{32}	−3658829788	c_{48}	−753564570		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A302370 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{63} - c_1 x^{62} - \dots - c_{63}$. Its constant term is $-c_{63} = -4956$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 63 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 63, so $\deg q = 63$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 63, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 63, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4956.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-63 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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